Oil Price Shocks and Texas Employment

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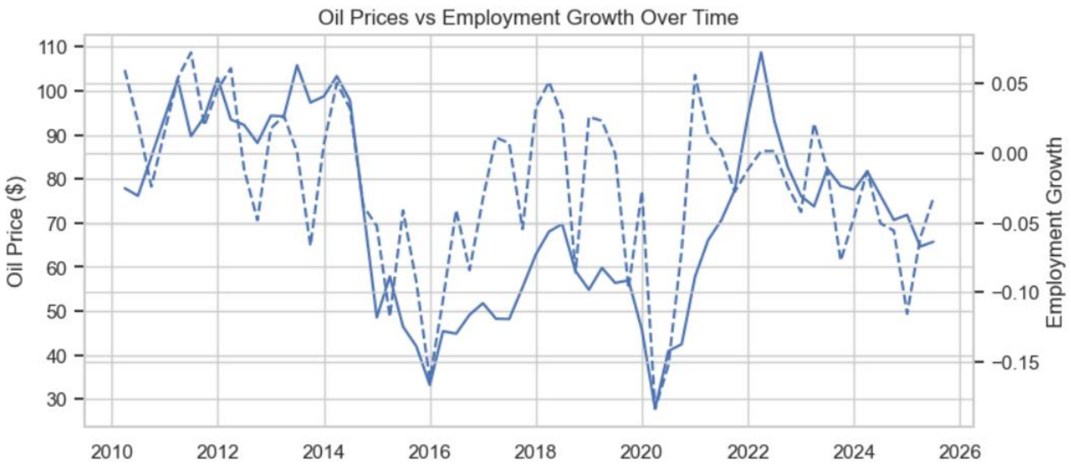


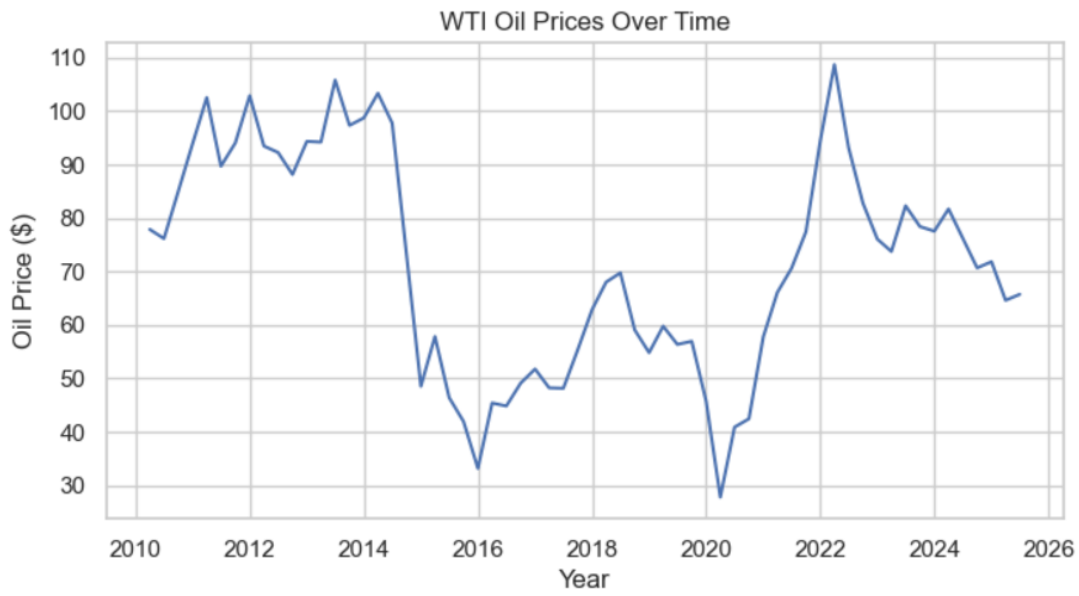
Hook

- WTI oil prices fell to about **$27.81** during the 2020 shock
- They later climbed above **$108**

Research question: Do oil price shocks actually predict Texas energy jobs?

Interpretation:

- If oil prices strongly controlled employment, Texas energy jobs should rise and fall sharply with price swings.
- Our project tests whether that assumption is actually true.



Research Question and Data

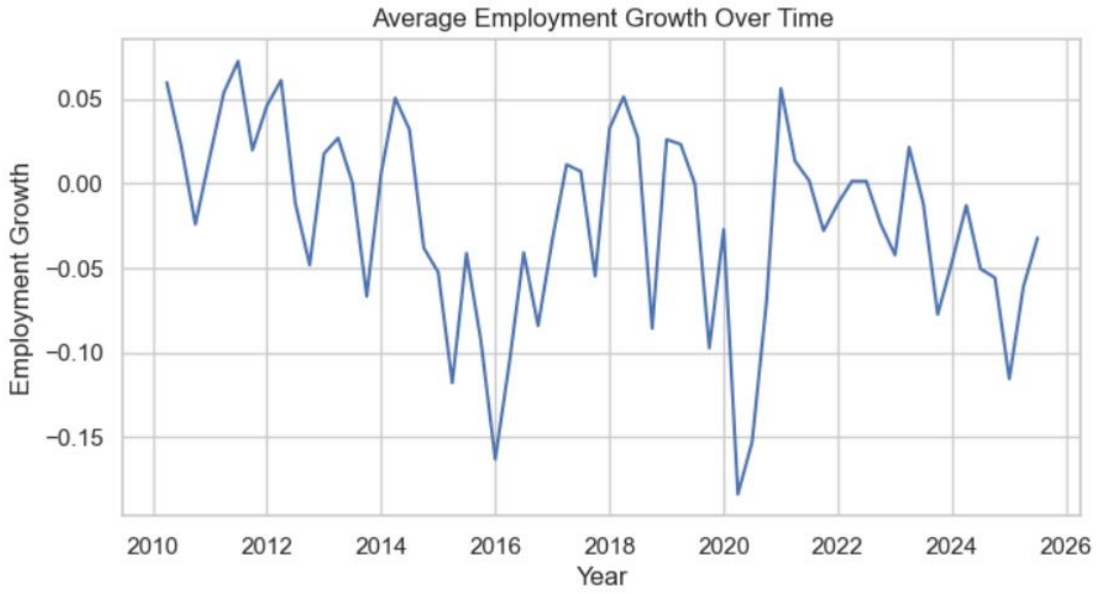
Research Question: Do changes in WTI oil prices predict county-level employment growth in Texas energy-related industries?

Data used:

- **FRED:** WTI oil prices and Texas unemployment rate
- **BLS QCEW:** county-level employment
- **2010–2025:** quarterly county-level observations
- **Final dataset:** 9,295 observations

Interpretation:

- This gives us both a statewide oil market signal and local employment outcomes.
- The goal is not just to see if oil prices move, but whether they translate into real job growth.



Key Variables

Interpretation:

- We expected oil prices to have a positive effect.
- But employment may also depend on hiring delays, technology, firm decisions, and local conditions.

Variable	Source	Expected Direction	Why It Matters
Employment Growth	BLS QCEW	Outcome	Measures job changes by county
WTI Oil Price	FRED	+	Higher oil prices may lead to more drilling, production, and hiring
Texas Unemployment	FRED	–	Controls for broader labor market weakness
Post-2020 Shock	Created variable	?	Tests whether the oil-job relationship changed after the shock

What the Data Looks Like

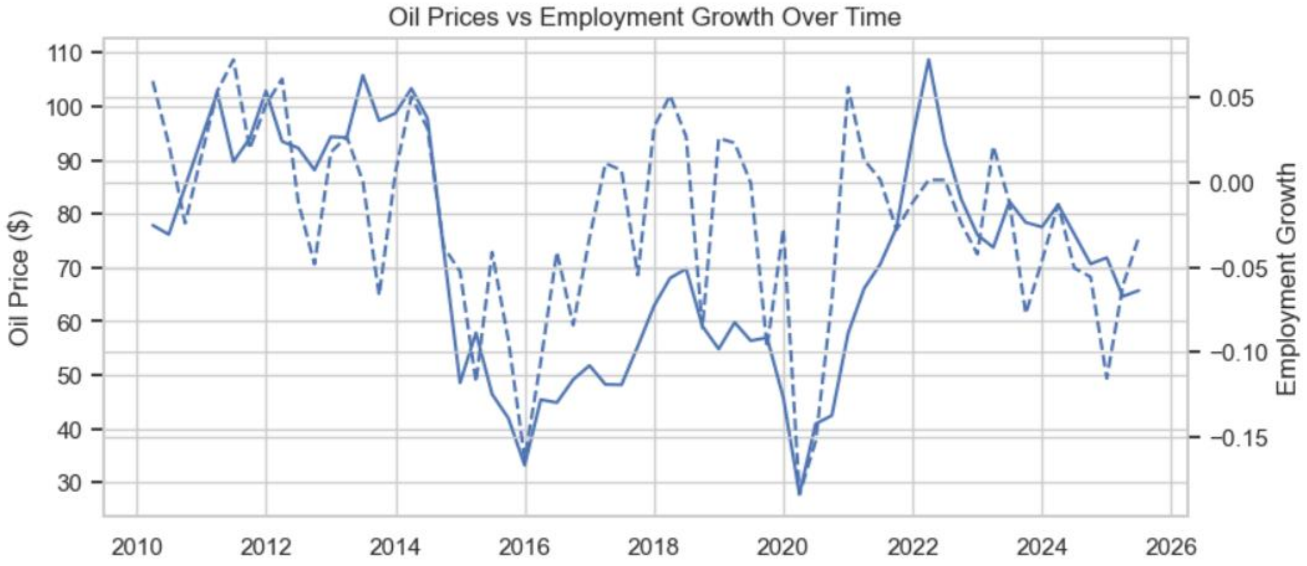
- Use one large time-series chart
- Show WTI oil price and average employment growth over time

Main takeaways:

- Oil prices are highly volatile.
- Employment growth is much more stable.
- Big oil swings happened in 2014–2016, 2020, and 2022.
- Employment did not move one-for-one with oil prices.

Interpretation:

- The visual already suggests a weak relationship.
- Oil prices clearly shock the market, but jobs appear slower and less reactive.



Our Approach

- **Model equation:**
  Employment Growth = $\beta_0 + \beta_1$(WTI Oil Price) + β_2(Unemployment) + ε

Methods used:

- **OLS regression:** tests whether oil prices predict employment growth
- **Regression with controls:** adds unemployment
- **Difference-in-Differences:** tests whether the relationship changed after 2020

Interpretation:

- OLS tells us the average relationship.
- The DiD model checks whether the 2020 shock changed that relationship.

Main Results

- **Main finding:**
  Oil prices are statistically significant, but weak as a predictor.

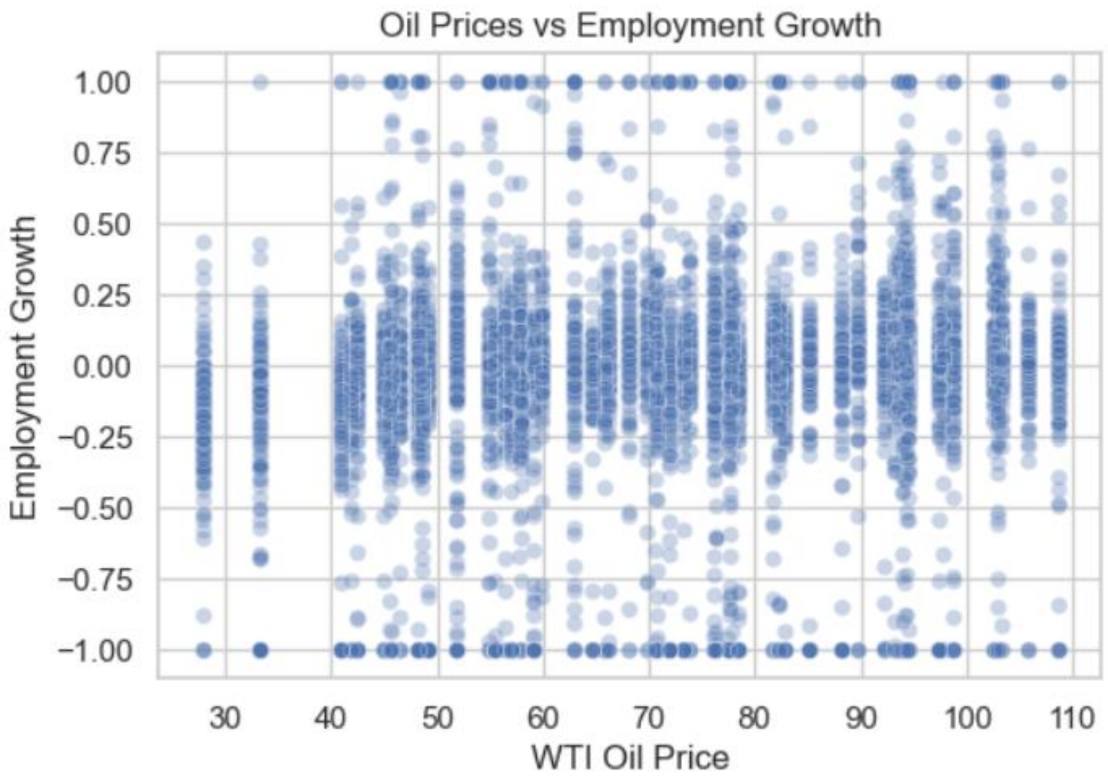
Result	Value	Meaning
Oil price coefficient	0.0016	\$1 increase in oil price → about 0.16 percentage point higher employment growth
P-value	0.000	Statistically significant
R^2	0.014	Oil prices explain only 1.4% of employment variation

Interpretation:

- The sign matches our expectation: higher oil prices are linked to higher employment growth.
- But the effect is small.
- A \$10 increase in oil prices predicts only about a 1.6 percentage point increase in employment growth.
- The low R^2 means oil prices alone do not explain most job changes.

Bottom line: Oil prices matter statistically, but they are not a strong practical predictor.

What the Model Gets Right and Wrong



What it gets right:

- Finds the expected positive relationship.
- Confirms oil prices have some connection to employment.

What it gets wrong:

- The model leaves about **98.6%** of employment variation unexplained.
- The scatterplot shows points spread widely, not tightly around a clear trend.
- It struggles during major shocks because employment depends on more than price.

Why the model struggles:

- Hiring may respond with a delay.
- Counties differ widely.
- Firms may adjust hours, investment, or technology before cutting jobs.
- Rig counts, production levels, wages, and company budgets are missing.

Robustness Check

DiD result:
Did the oil-employment relationship change after 2020?

Result	Value	Meaning
Post-2020 coefficient	−0.0315	Employment growth was lower after 2020 overall
Interaction coefficient	0.0001	Almost no change in the oil price effect
Interaction p-value	0.722	Not statistically significant
R^2	0.0156	Still very low explanatory power

Interpretation:

- The 2020 shock did not significantly change the relationship.
- Even after a historic oil market disruption, oil prices still remained a weak predictor of employment.

Implications

Audience	Key Takeaway
Energy companies	Do not base hiring plans on oil prices alone
Investors	Oil price is a weak signal for employment trends
Policymakers	Job growth depends on broader local and industry factors
Analysts	Add rig counts, production, wages, and county fixed effects

Interpretation:

- WTI oil prices are useful, but incomplete.
- A single commodity price doesn’t tell the whole story.

Limitations and Next Steps

Limitations:

- **Missing variables:** rig counts, production, wages, and firm-level decisions are not included.
- **Timing issue:** employment may respond months or quarters after oil prices change.
- **Local variation:** Texas counties may react differently depending on their energy exposure.

Next step: Add lagged oil prices, rig count data, and county fixed effects to better capture local employment responses.

Conclusion

- **Answer:** Oil prices predict Texas energy employment growth in the expected positive direction, but the relationship is weak.
- **Meaning:** Oil prices matter, but they are not enough to explain or forecast Texas energy job growth.
- **Final takeaway:** The biggest finding is not that oil prices do not matter; it is that they matter less than expected.

Thank you

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GitHub: https://github.com/LoganAverill/BANA-4373-Final-Project---Bauer-Daniel-Averill